Module 5 Homework: Three Stock Research Ideas

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Three exploratory research ideas that differ in economic mechanism, data source and horizon: a demand signal (Indian auto OEMs), a relative-value strategy (private bank pairs) and text-based information processing (earnings-call language in less-covered stocks). All data sources are labeled "to verify" until opened and checked.

Question 1: Retail registrations vs. company dispatches in Indian auto OEMs

Market and scope. NSE-listed two-wheeler, passenger vehicle and tractor makers (Hero MotoCorp, Bajaj Auto, TVS Motor, Maruti Suzuki, M&M, Eicher). A possible extension is listed auto-ancillary mid-caps. The horizon is 1–3 months.

Investment thesis. Companies report wholesale dispatches (sales to dealers), while actual consumer demand shows up in retail registrations. If dispatches run well ahead of registrations for several months, dealer inventory is building. That tends to be followed by production cuts, discounting and weaker margins. If registrations outpace dispatches, inventory is lean and upside surprises become more likely.

Potential edge. Most investors react to headline monthly sales numbers. The registration-to-dispatch gap needs aggregation work, such as mapping Vahan's maker and class categories to listed companies, and that tedious step may leave the signal partly unpriced. Limitation: large caps are heavily covered by sell-side analysts, so the edge may be small there. Testing it on less-covered names is part of the research question.

Data and sources (all to verify).

- Vahan dashboard (Ministry of Road Transport & Highways): registrations by maker and vehicle class. Check the history depth, whether data can be exported or scraped, and how revisions are handled.
- Monthly dispatch figures from company press releases and stock exchange filings, and SIAM or FADA monthly releases. FADA also publishes retail and inventory commentary. Check publication dates.
- NSE daily price data for returns.

Research approach. For each company and month, compute inventory change ≈ dispatches − registrations, normalized by trailing volume. Rank firms cross-sectionally each month and go long the lowest-inventory tercile and short (or underweight) the

highest. Test forward returns over 1, 2 and 3 months, using only information available on the date it was published.

Validation and risks.

- Benchmark: equal-weight basket of the same stocks, plus Nifty Auto.
- Reject the idea if the signal's return spread disappears after controlling for momentum and last month's sales growth, or if the sign flips across sub-periods.
- Risks: registrations lag sales by days to weeks, and festival timing distorts months. Exports and commercial vehicles don't map cleanly. The sample is small, with only about 6–10 firms. Shorting single stocks is limited to F&O names.

Question 2: Pairs trading among Indian private banks

Market and scope. F&O-eligible private-sector banks (for example ICICI Bank and Axis Bank, or Kotak and IndusInd, with other candidate pairs screened). The horizon is days to a few weeks.

Investment thesis. Banks in the same segment share exposure to interest rates, credit cycles, regulation and liquidity. Their relative price should be mean-reverting. Temporary divergences caused by flows, such as FPI selling, index rebalances or block deals, carry no new information about fundamentals and should close.

Potential edge. The edge is practical rather than informational. In India, overnight short selling in cash equities is not available to most retail participants, so pairs have to be built with single-stock futures. That limits who can run the strategy, and the main constraint becomes margin and costs. The research contribution is distinguishing flow-driven divergence from fundamental divergence.

Data and sources (to verify).

- NSE bhavcopy or an adjusted price series, with futures prices and open interest.
- Quarterly results and RBI announcements to flag fundamental events.
- Corporate action history for adjustments, and FPI flow data.

Research approach.

- Run an Engle–Granger cointegration test, or fit a rolling-window hedge ratio.
- Compute the spread z-score and enter at ±2.
- Exit at 0 and use a stop-loss at about ±3.5.
- Skip signals near earnings or major company news, where the gap is likely informational.
- Use rolling re-estimation so every test is out of sample.

Validation and risks.

- Benchmarks: a naive long-short with no filters, and a buy-and-hold basket.
- Reject the idea if the relationship doesn't hold out of sample, if returns vanish after STT, brokerage, slippage and futures roll costs, or if a few trades drive all the profit.
- Risks: structural breaks (the 2023 HDFC Bank and HDFC Ltd merger is an example of how a pair relationship can permanently change), regime shifts, and crowding.

Question 3: Post-earnings drift from earnings-call language in less-covered Indian mid- and small-caps

Market and scope. NSE/BSE-listed companies with a market cap in a mid-to-small range and few analysts, in sectors where management guidance matters (such as specialty chemicals, capital goods and consumer). The horizon is 1–3 months after each quarterly result.

Investment thesis. Reported numbers are reflected quickly, but forward-looking content in call transcripts and presentations, such as guidance, capacity plans, order book commentary and tone, is slower to be digested. For thinly covered stocks, the information may spread slowly, leading to drift in the direction of the language change.

Potential edge. Few analysts cover these companies, and reading transcripts at scale takes effort. A text pipeline can systematically score language across hundreds of companies each earnings season. The real question is whether the text carries information beyond the earnings surprise and the initial price reaction.

Data and sources (to verify).

- Earnings-call transcripts: listed companies are, as far as I know, required to publish them on their websites and on the exchanges. Check the actual rules, how consistently companies comply, and the delay.
- BSE/NSE corporate announcement feeds for results and investor presentations.
- Price data, and consensus estimates if accessible (likely limited).
- Check the file formats first, since many are PDFs.

Research approach. Extract management language on guidance, demand and costs. Start simple, with a keyword or lexicon score and the change versus the prior quarter's call. Move to an LLM-based score only if the baseline shows promise. Sort stocks into quintiles by the score change and measure returns from day 2 after the call, so that the initial reaction is excluded.

Validation and risks.

- Benchmarks: a basket sorted by earnings surprise alone (YoY profit growth) and one sorted by the day-0/1 price reaction. The language score must add something beyond both.
- Reject the idea if the score adds nothing after those controls or the effect only appears in the most illiquid names.
- Risks: poor transcript availability, publication delays, wide spreads and low liquidity in small caps, look-ahead bias, and LLM scores that are unstable across runs.

Before submitting

Open Vahan, the NSE bhavcopy page and a handful of transcripts so you can honestly state which sources you have checked; the assignment grades that distinction. Add one concrete first test per idea, such as "pull 24 months of Vahan data for 6 OEMs and compute the gap." Verify all factual claims and adjust the ideas so they reflect your own judgment.